



Fundamentals of Data Engineering / BigData

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C-CAT section B
big Data -> 5 MCQ => theory

next 3 days

Quizes

Q1: 10 mcq => day 3 => 26 june

Q2: 10 mcq => day 4 => 27 june

End : 20 mcq => day 5 => 28 june

Mock => 10

@daily => 20

@ 70 MCQ

DAC => DBDA

pre-requistes

1: appli dev => programming

2: database

3: networking

4: OS -> linux



- **Big Data Fundamentals**

- Evolution of Data Engineering | V's: Volume, Velocity, Variety, Veracity, Value

day1

- **Databases**

- RDBMS - ACID, SQL (basic concept only) | NoSQL - BASE, CAP theorem

- **Data warehouse - OLAP vs OLTP**

- Data cleansing, Data transformations and Data modelling | Data warehouse vs Data mart

- **Data Engineering Life Cycle**

- Source → Ingestion → Storage → Transformation → Serving
- Ingestion: ETL vs ELT
- Storage: Distributed storage, Storage services | Processing: Batch vs Stream

day2

- **Cloud computing fundamentals**

- Virtualization, Scaling, Elasticity, Cloud service models, Vendors

- **Big Data Technologies**

- Frameworks: Hadoop, Hive, Spark, Kafka
- Applications and Job profiles.

day3



Data Engineering at a Glance



be4 1970 file i/o
Database & Warehouse
@1970
RDBMS
CRUD

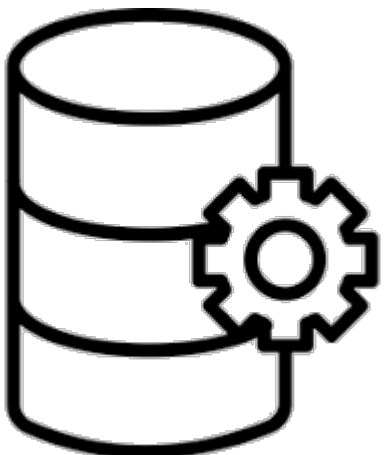
DWH @1990



@1980
Internet & DotCom
1995 -> java applet
Data burst
@2000



NoSQL Database
@1998
CRUD
4+2



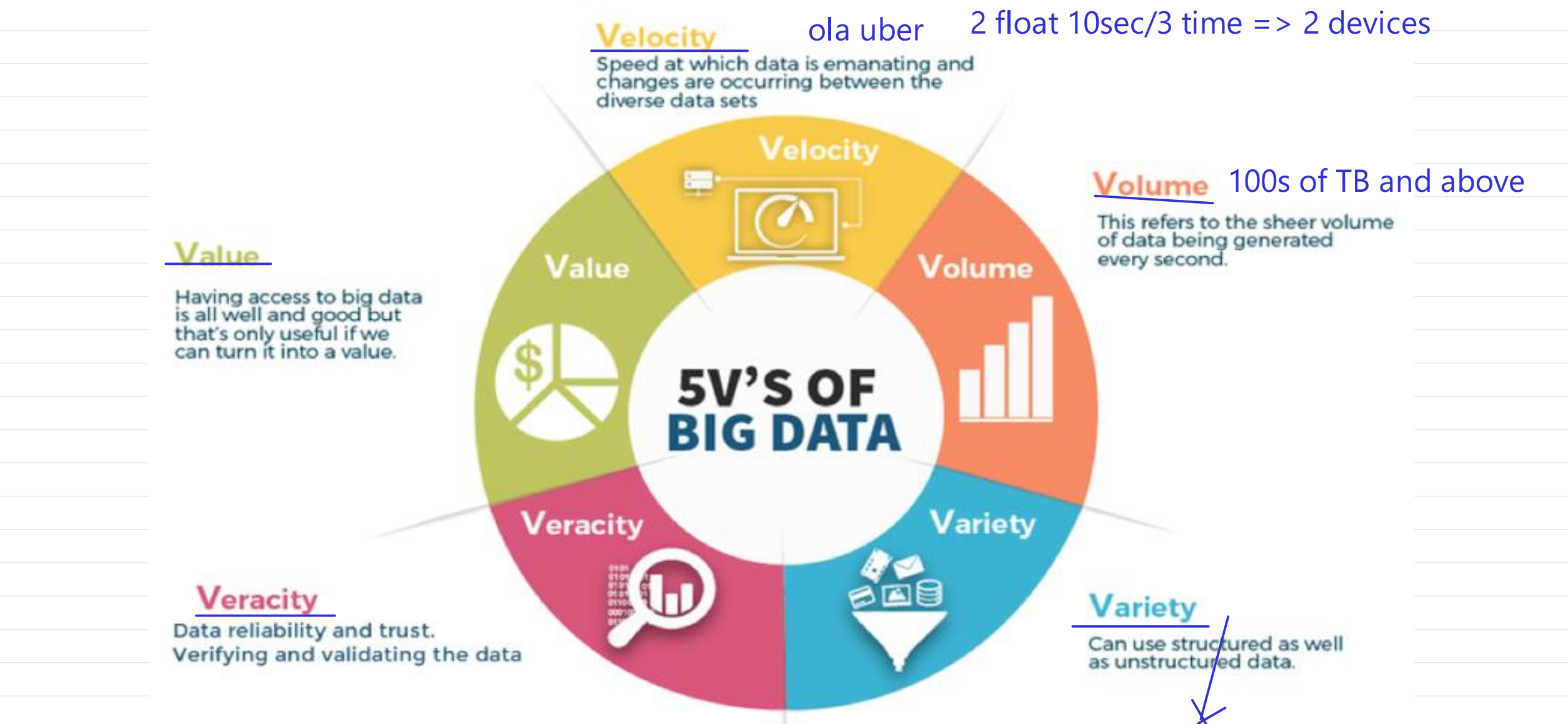
massive parallel processing
MPP & Big Data Tech
@2000
i3 +4GB => 100
@2003-4 => google



rent
Cloud Computing
@2010



Big Data characteristics



Types of Data

facebook 2008

```
{  
  post:"--",  
  links:--,  
  writer:--,  
  img:jpg  
  videos:--,  
  loc:--,  
  tag:--  
  react:--}
```



STRUCTURED DATA

Uses pre-defined data models filled with labels, numbers and values.



Excel spreadsheets,
electronic forms,
data tables

—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—

fixed
format /
schema



SEMI-STRUCTURED DATA

Mainly unstructured but uses internal tags and markings to help classify.



Email stores, JSON,
NoSql, XML

flexible schema



UNSTRUCTURED DATA

No pre-defined data model; packed with text and information.



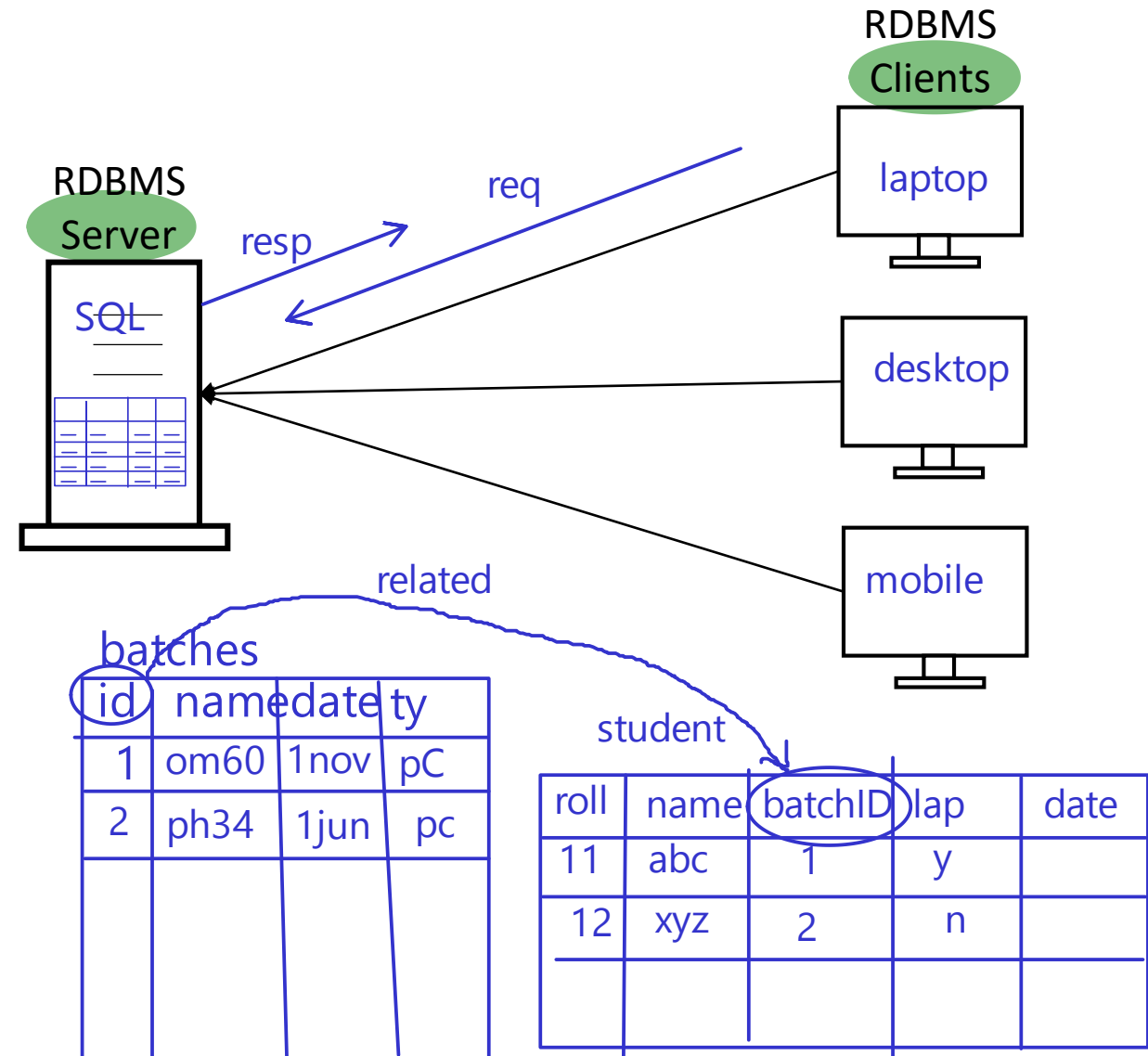
Scanned PDFs, text
documents, audio
or video files

ML



RDBMS relational DBMS => CRUD

- Every enterprise application need to manage data.
- RDBMS is relational DBMS than manages **structured data.**
- Data is organized into tables, rows and columns. Tables are related to each other.
- All enterprise RDBMS follow server-client architecture, have built-in relational capabilities, fully ACID **transactions**, based on Codd's rules.
- DB2, Oracle, MS-SQL, MySQL, Postgre-SQL, MS-Access, SQLite, etc.



- RDBMS data is processed with SQL queries.
- ANSI standardised in 1986 and ISO Standardization in 1987.
- Five major categories:
 - DDL: Data Definition Language e.g. CREATE, ALTER, DROP, RENAME.
 - CREATE TABLE student(roll INT, name CHAR(40), batchid INT);
 - DML: Data Manipulation Language e.g. INSERT, UPDATE, DELETE.
 - INSERT INTO student VALUES(1, 'Ravi', 3);
 - UPDATE student SET name='Ravee' WHERE roll=1;
 - DELETE FROM student WHERE roll=1;
 - DQL: Data Query Language e.g. SELECT.
 - SELECT * FROM student;
 - DCL: Data Control Language e.g. CREATE USER, GRANT, REVOKE.
 - TCL: Transaction Control Language e.g. SAVEPOINT, COMMIT, ROLLBACK.



Transaction characteristics- ACID

accounts

id	type	balance
1	saving	30000
2	saving	5000
3	saving	80000
4	current	40000

4000

accounts

id	type	balance
1	saving	30000 26000
2	saving	5000 9000
3	saving	80000
4	current	40000

$30000 - 4000 = 26000 \Rightarrow$ update on acc1 ✓

$5000 + 4000 = 9000 \Rightarrow$ update on acc2 ✗

Transaction = set of DML commands executed as a single unit $\rightarrow$ atomic

consistant $\Rightarrow$ same result shown to all the clients

isolated $\Rightarrow$ several trasactions executed at a time without affecting each other

Durable $\rightarrow$ all changes are saved permanantly



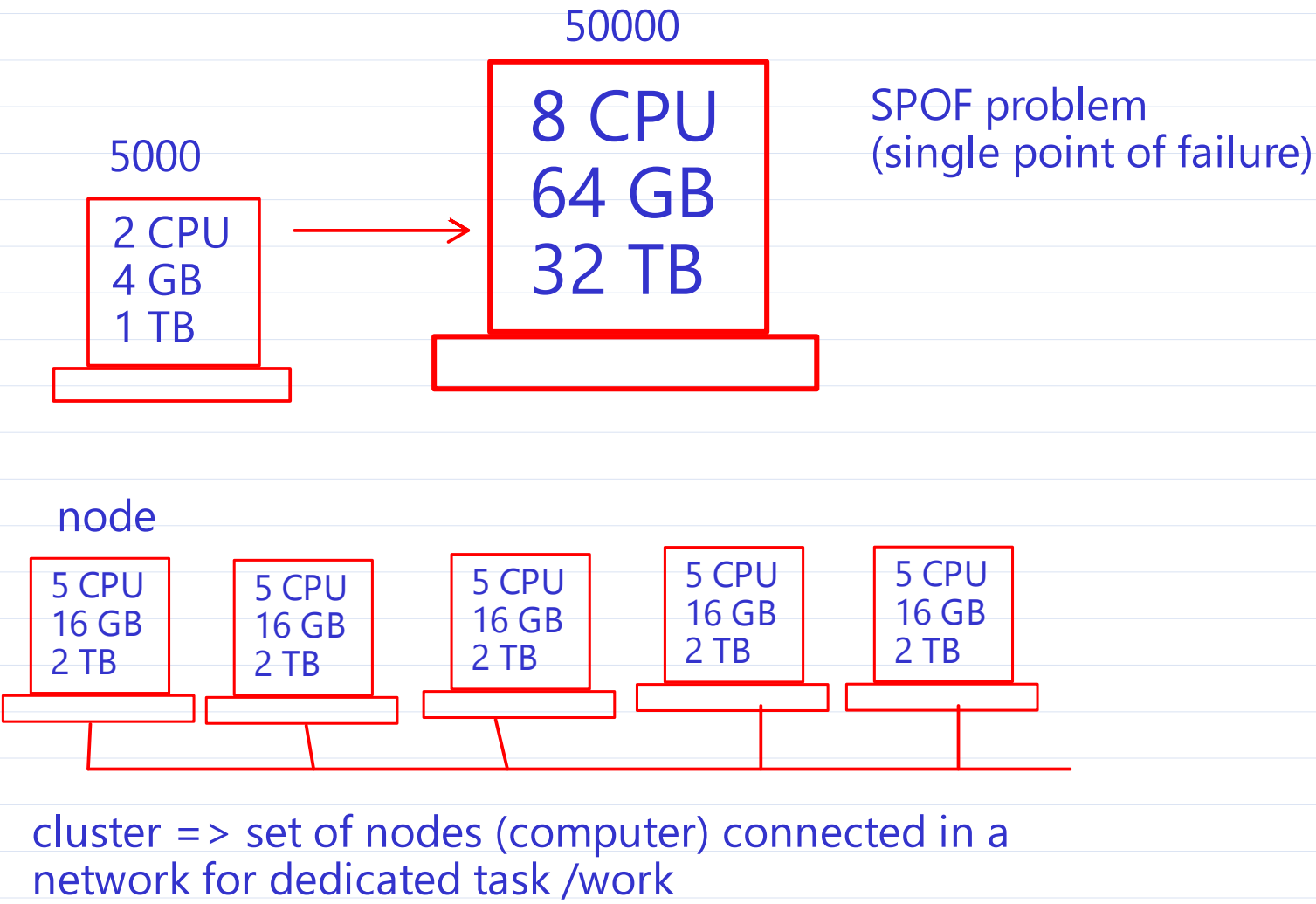
Scalability

clients
data
load
work

Vertical scaling
[up - scaling]

Horizontal scaling
[out - scaling]

[distributed computing]



- Scalability is the ability of a system to expand to meet your business needs.
- Scalability describes a elasticity of the system, ability to adapt to change and demand.
- Good scalability ensures the quality of your service.

Vertical Scaling



CPU: 3 , RAM: 6G



CPU: 2 , RAM: 4G



CPU: 1 , RAM: 2G



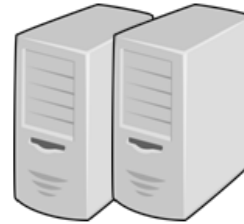
Vertical scaling describes adding more resources to your current machines.

- ✓ increase memory in the system
- ✓ expanding storage by adding hard drives
- ✓ upgrading the CPUs.
- ✓ upgrading network speed.

Horizontal Scaling



1 PC (CPU: 1, RAM:2G)



2 PC (CPU: 1, RAM:2G)



3 PC (CPU: 1, RAM:2G)



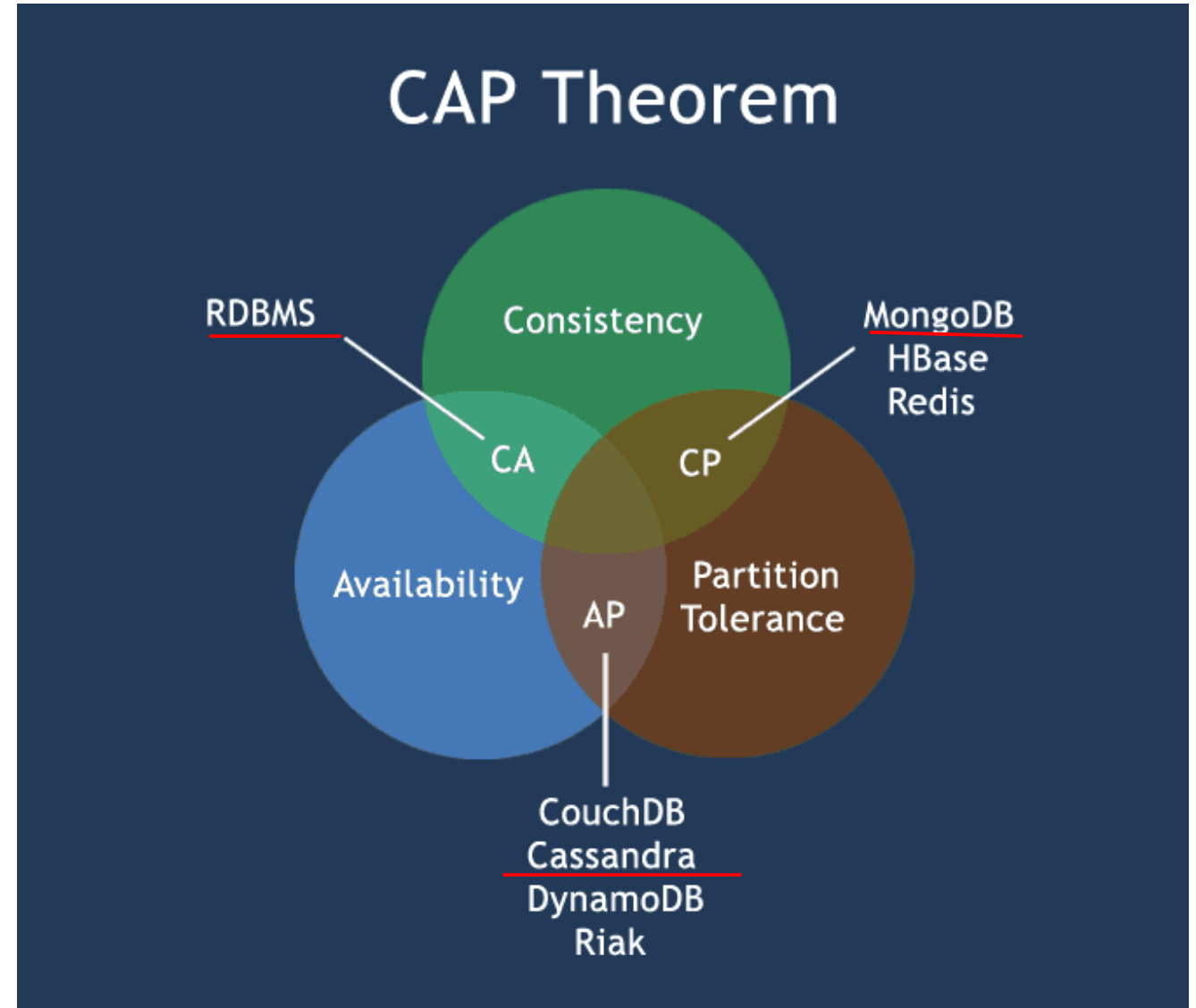
Horizontal Scaling refers to adding additional nodes or machines to your infrastructure to cope with new demands.

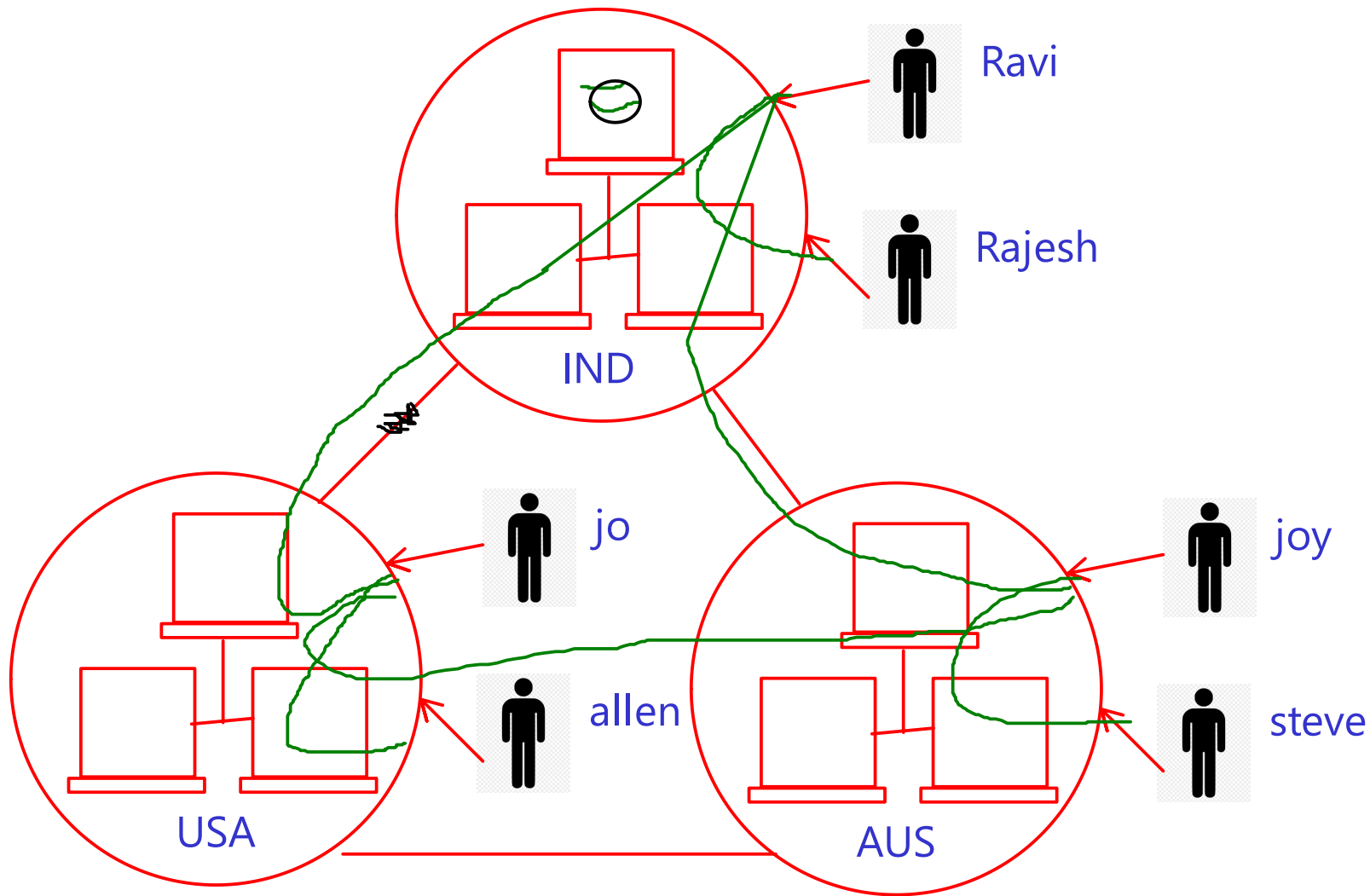
- ✓ adding a new computer to a distributed software application

- Stands for Not Only SQL
- Manages structured and semi-structured data.
- Prioritizes high performance, high availability and scalability
- Designed for Horizontal scaling. Reliable, fault tolerant, Better performance/Speed.
- No declarative query language
- Uses: Huge data (TBs), Many Read/Write ops, Scalable, Flexible schema.
- Don't use if: Need high consistency, Multiple relations
- BASE transactions and Based on CAP Theorem



- **Consistency** - Data is consistent after operation. After an update operation, all clients see the same data.
- **Availability** - System is always on (i.e. service guarantee), no downtime. 24X7
- **Partition Tolerance** - System continues to function even the communication among the servers is unreliable.





BASE transaction

BA => Basically available
24 X 7 all time available

S => soft state

data is auto transferred
to all nodes in the cluster

E => eventual consistency

same data visible to
all the client eventually
(after some time
not immediately)

MCQ

Q: 1. ROLLBACK is _____ type command .

- A. DCL
- B. TCL
- C. DDL
- D. DDD



MCQ

Q: 1. ROLLBACK is _____ type command .

A. DCL

B. TCL

C. DDL

D. DDD



MCQ

Q: 1. _____ command is used to delete table.

- A. FREE
- B. DELEET
- C. DROP
- D. RELISED



MCQ

Q: 1. _____ command is used to delete table.

A. FREE

B. DELEET

C. DROP

D. RELISED



Q: 1. Which one is not in v's of Big Data?

- A. Variety
- B. Velocity
- C. Volatile
- D. volume

Q: 1. Which one is not in v's of Big Data?

- A. Variety -> Data can be unstructured, semi-structured or structured
- B. Velocity -> Data generated with high speed
- C. Volatile
- D. volume -> Huge amount of data

Veracity in Big data means_____.

- A. The data is generated with high speed
- B. The data is huge
- C. The data is reliable and trustworthy
- D. The data management

Veracity in Big data means_____.

- A. The data is generated with high speed -> Velocity
- B. The data is huge -> volume
- C. The data is reliable and trustworthy -> Veracity
- D. The data management -> database

In RDBMS data is stored in _____.

- A. document
- B. tables
- C. collection
- D. keys



In RDBMS data is stored in _____.

A. document

B. tables

C. collection

D. keys



_____ introduced the NoSQL concept in 1998.

- A. Cassandra
- B. Devid Sam
- C. Carl Strozz
- D. E.F. CODD



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A. Cassandra

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Thank you!

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